Precalculus — Algebra I

Version 4 $- \epsilon$

by

Carl Stitz, Ph.D.
Lakeland Community College

Jeff Zeager, Ph.D.
Lorain County Community College

$\sigma o \phi \acute{\imath} \alpha$

© 2025 Carl Stitz and Jeff Zeager
https://stitz-zeager.com/

This work is licensed under CC BY-NC-SA 3.0. To view a copy of this licence, visit:

https://creativecommons.org/licenses/by-nc-sa/3.0/deed.en

Sophia Publishing
CHENNAI
https://sophia-publishing.github.io/

Table of Contents

Preface **v**

1 Introduction to Functions **1**
1.1 Functions and their Representations 1
1.2 Constant and Linear Functions 2
1.3 Absolute Value Functions 3
1.4 Quadratic Functions 4

2 Polynomial Functions **5**
2.1 Graphs of Polynomial Functions 5
2.2 The Remainder and Factor Theorems 6
2.3 Real Zeros of Polynomials 7
2.4 Complex Zeros and the Fundamental Theorem of Algebra 8

3 Rational Functions **9**
3.1 Introduction to Rational Functions 9
3.2 Graphs of Rational Functions 10
3.3 Inequalities involving Rational Functions and Applications 11
3.3.1 Exercises 24
3.3.2 Answers 28

Preface

Thank you for your interest in our book, but more importantly, thank you for taking the time to read the Preface. I always read the Prefaces of the textbooks which I use in my classes because I believe it is in the Preface where I begin to understand the authors - who they are, what their motivation for writing the book was, and what they hope the reader will get out of reading the text. Pedagogical issues such as content organization and how professors and students should best use a book can usually be gleaned out of its Table of Contents, but the reasons behind the choices authors make should be shared in the Preface. Also, I feel that the Preface of a textbook should demonstrate the authors' love of their discipline and passion for teaching, so that I come away believing that they really want to help students and not just make money. Thus, I thank my fellow Preface-readers again for giving me the opportunity to share with you the need and vision which guided the creation of this book and passion which both Carl and I hold for Mathematics and the teaching of it.

Carl and I are natives of Northeast Ohio. We met in graduate school at Kent State University in 1997. I finished my Ph.D in Pure Mathematics in August 1998 and started teaching at Lorain County Community College in Elyria, Ohio just two days after graduation. Carl earned his Ph.D in Pure Mathematics in August 2000 and started teaching at Lakeland Community College in Kirtland, Ohio that same month. Our schools are fairly similar in size and mission and each serves a similar population of students. The students range in age from about 16 (Ohio has a Post-Secondary Enrollment Option program which allows high school students to take college

courses for free while still in high school.) to over 65. Many of the "non-traditional" students are returning to school in order to change careers. A majority of the students at both schools receive some sort of financial aid, be it scholarships from the schools' foundations, state-funded grants or federal financial aid like student loans, and many of them have lives busied by family and job demands. Some will be taking their Associate degrees and entering (or re-entering) the workforce while others will be continuing on to a four-year college or university. Despite their many differences, our students share one common attribute: they do not want to spend $200 on a College Algebra book.

The challenge of reducing the cost of textbooks is one that many states, including Ohio, are taking quite seriously. Indeed, state-level leaders have started to work with faculty from several of the colleges and universities in Ohio and with the major publishers as well. That process will take considerable time so Carl and I came up with a plan of our own. We decided that the best way to help our students right now was to write our own College Algebra book and give it away electronically for free. We were granted sabbaticals from our respective institutions for the Spring semester of 2009 and actually began writing the textbook on December 16, 2008. Using an open-source text editor called TexNicCenter and an open-source distribution of LaTeX called MikTex 2.7, Carl and I wrote and edited all of the text, exercises and answers and created all of the graphs (using Metapost within LaTeX) for Version $0.\overline{9}$ in about eight months. (We choose to create a text in only black and white to keep printing costs to a minimum for those students who prefer a printed edition. This somewhat Spartan page layout stands in sharp relief to the explosion of colors found in most other College Algebra texts, but neither Carl nor I believe the four-color print adds anything of value.) I used the book in three sections of College Algebra at Lorain County Community College in the Fall of 2009 and Carl's colleague, Dr. Bill Previts, taught a section of College Algebra at Lakeland with the book that semester as well. Students had the option of downloading the book as a .pdf file from our website www.stitz-zeager.com[1] or buying a low-cost printed version from our colleges' respective bookstores. (By

[1] http://www.stitz-zeager.com

giving this book away for free electronically, we end the cycle of new editions appearing every 18 months to curtail the used book market.) During Thanksgiving break in November 2009, many additional exercises written by Dr. Previts were added and the typographical errors found by our students and others were corrected. On December 10, 2009, Version $\sqrt{2}$ was released. The book remains free for download at our website and by using Lulu.com[2] as an on-demand printing service, our bookstores are now able to provide a printed edition for just under $19. Neither Carl nor I have, or will ever, receive any royalties from the printed editions. As a contribution back to the open-source community, all of the LaTeX files used to compile the book are available for free under a Creative Commons License on our website as well. That way, anyone who would like to rearrange or edit the content for their classes can do so as long as it remains free.

The only disadvantage to not working for a publisher is that we don't have a paid editorial staff. What we have instead, beyond ourselves, is friends, colleagues and unknown people in the open-source community who alert us to errors they find as they read the textbook. What we gain in not having to report to a publisher so dramatically outweighs the lack of the paid staff that we have turned down every offer to publish our book. (As of the writing of this Preface, we've had three offers.) By maintaining this book by ourselves, Carl and I retain all creative control and keep the book our own. We control the organization, depth and rigor of the content which means we can resist the pressure to diminish the rigor and homogenize the content so as to appeal to a mass market. A casual glance through the Table of Contents of most of the major publishers' College Algebra books reveals nearly isomorphic content in both order and depth. Our Table of Contents shows a different approach, one that might be labeled "Functions First." To truly use The Rule of Four, that is, in order to discuss each new concept algebraically, graphically, numerically and verbally, it seems completely obvious to us that one would need to introduce functions first. (Take a moment and compare our ordering to the classic "equations first, then the Cartesian Plane and THEN functions" approach seen in most of

[2] http://www.lulu.com/content/paperback-book/college-algebra/7513097

the major players.) We then introduce a class of functions and discuss the equations, inequalities (with a heavy emphasis on sign diagrams) and applications which involve functions in that class. The material is presented at a level that definitely prepares a student for Calculus while giving them relevant Mathematics which can be used in other classes as well. Graphing calculators are used sparingly and only as a tool to enhance the Mathematics, not to replace it. The answers to nearly all of the computational homework exercises are given in the text and we have gone to great lengths to write some very thought provoking discussion questions whose answers are not given. One will notice that our exercise sets are much shorter than the traditional sets of nearly 100 "drill and kill" questions which build skill devoid of understanding. Our experience has been that students can do about 15-20 homework exercises a night so we very carefully chose smaller sets of questions which cover all of the necessary skills and get the students thinking more deeply about the Mathematics involved.

Critics of the Open Educational Resource movement might quip that "open-source is where bad content goes to die," to which I say this: take a serious look at what we offer our students. Look through a few sections to see if what we've written is bad content in your opinion. I see this open-source book not as something which is "free and worth every penny", but rather, as a high quality alternative to the business as usual of the textbook industry and I hope that you agree. If you have any comments, questions or concerns please feel free to contact me at jeff@stitz-zeager.com or Carl at carl@stitz-zeager.com.

Jeff Zeager
Lorain County Community College
January 25, 2010

Chapter 1

Introduction to Functions

1.1 Functions and their Representations

1.2 Constant and Linear Functions

1.3 Absolute Value Functions

1.4 Quadratic Functions

Chapter 2

Polynomial Functions

2.1 Graphs of Polynomial Functions

2.2 The Remainder and Factor Theorems

2.3 Real Zeros of Polynomials

2.4 Complex Zeros and the Fundamental Theorem of Algebra

Chapter 3

Rational Functions

3.1 Introduction to Rational Functions

3.2 Graphs of Rational Functions

3.3 Inequalities involving Rational Functions and Applications

In this section, we solve equations and inequalities involving rational functions and explore associated application problems. Our first example showcases the critical difference in procedure between solving equations and inequalities.

Example 3.3.1.

1. Solve $\dfrac{x^3 - 2x + 1}{x - 1} = \dfrac{1}{2}x - 1$.
2. Solve $\dfrac{x^3 - 2x + 1}{x - 1} \geq \dfrac{1}{2}x - 1$.
3. Verify your solutions using a graphing utility.

Solution.

1. To solve the equation, we clear denominators

$$\begin{aligned}
\frac{x^3 - 2x + 1}{x - 1} &= \frac{1}{2}x - 1 \\
\left(\frac{x^3 - 2x + 1}{x - 1}\right) \cdot 2(x - 1) &= \left(\frac{1}{2}x - 1\right) \cdot 2(x - 1) \\
2x^3 - 4x + 2 &= x^2 - 3x + 2 && \text{(expand)} \\
2x^3 - x^2 - x &= 0 \\
x(2x + 1)(x - 1) &= 0 && \text{(factor)} \\
x &= -\frac{1}{2},\ 0,\ 1
\end{aligned}$$

Since we cleared denominators, we need to check for extraneous solutions. Sure enough, we see that $x = 1$ does not satisfy the original equation, so our only solutions are $x = -\frac{1}{2}$ and $x = 0$.

2. To solve the inequality, it may be tempting to begin as we did with the equation – namely by multiplying both sides by the quantity $(x-1)$. The problem is that, depending on x, $(x-1)$ may be positive (which doesn't affect the inequality) or $(x-1)$ could be negative (which would reverse the inequality). Instead of working by cases, we collect all of the terms on one side of the inequality with 0 on the other and make a sign diagram using the technique given on page **??** in Section **??**.

$$\frac{x^3-2x+1}{x-1} \geq \frac{1}{2}x-1$$

$$\frac{x^3-2x+1}{x-1} - \frac{1}{2}x+1 \geq 0$$

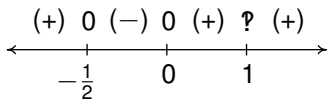
$$\frac{2\left(x^3-2x+1\right)}{2(x-1)} - \frac{x(x-1)}{2(x-1)} + \frac{2(x-1)}{2(x-1)} \geq 0$$

(get a common denominator)

$$\frac{2\left(x^3-2x+1\right)-x(x-1)+2(x-1)}{2(x-1)} \geq 0$$

$$\frac{2x^3-x^2-x}{2x-2} \geq 0 \qquad \text{(expand)}$$

Viewing the left hand side as a rational function $r(x)$ we make a sign diagram. The only value excluded from the domain of r is $x = 1$ which is the solution to $2x-2=0$. The zeros of r are the solutions to $2x^3-x^2-x=0$, which we have already found to be $x=0$, $x=-\frac{1}{2}$ and $x=1$, the latter was discounted as a zero because it is not in the domain. Choosing test values in each test interval, we construct the sign diagram below.



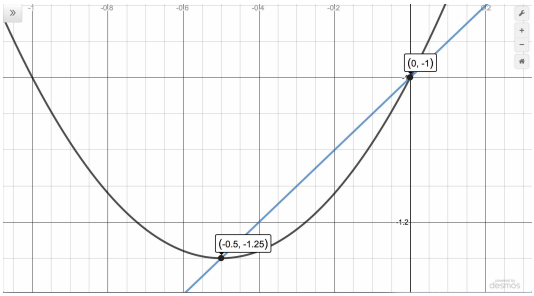


Figure 3.3.1

We are interested in where $r(x) \geq 0$. We see $r(x) > 0$, or $(+)$, on the intervals $\left(-\infty, -\frac{1}{2}\right)$, $(0, 1)$ and $(1, \infty)$. We know $r(x) = 0$ when $x = -\frac{1}{2}$ and $x = 0$. Hence, $r(x) \geq 0$ on $\left(-\infty, -\frac{1}{2}\right] \cup [0, 1) \cup (1, \infty)$.

3. To check our answers graphically, let $f(x) = \frac{x^3-2x+1}{x-1}$ and $g(x) = \frac{1}{2}x - 1$. The solutions to $f(x) = g(x)$ are the x-coordinates of the points where the graphs of $y = f(x)$ and $y = g(x)$ intersect. We graph both f and g in Figure 3.3.1 (the graph of g is the line and is slightly lighter in color.) We find only two intersection points, $(-0.5, -1.25)$ and $(0, -1)$ which correspond to our solutions $x = -\frac{1}{2}$ and $x = 0$. The solution to $f(x) \geq g(x)$ represents not only where the graphs meet, but the intervals over which the graph of $y = f(x)$ is above $(>)$ the graph of $g(x)$. From the graph, this *appears* to happen on on $\left(-\infty, -\frac{1}{2}\right] \cup [0, \infty)$ which *almost* matches the answer we found analytically. We have to remember that f is not defined at $x = 1$, so it cannot be included in our solution.[1] See Figure 3.3.2

□

The important take-away from Example 3.3.1 is not to clear fractions when working with an inequality unless you know for certain the sign of the denominators. We offer another example.

Example 3.3.2. Solve: $2t(3t-2)^{-1} \leq 3t^2(3t-2)^{-2}$. Check your answer

[1] We invite the reader to show there is a hole in the graph of $y = f(x)$ at $(1, 1)$.

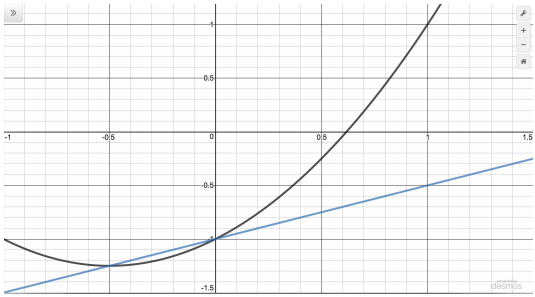


Figure 3.3.2

using a graphing utility.

Solution. We begin by rewriting the terns with negative exponents as fractions and gathering all nonzero terms to one side of the inequality:

$$
\begin{aligned}
2t(3t-2)^{-1} &\leq 3t^2(3t-2)^{-2} \\
\frac{2t}{3t-2} &\leq \frac{3t^2}{(3t-2)^2} \\
\frac{2t}{3t-2} - \frac{3t^2}{(3t-2)^2} &\leq 0 \\
\frac{2t(3t-2)}{(3t-2)^2} - \frac{3t^2}{(3t-2)^2} &\leq 0 && \text{(get a common denominator)} \\
\frac{2t(3t-2)-3t^2}{(3t-2)^2} &\leq 0 \\
\frac{3t^2-4t}{(3t-2)^2} &\leq 0 && \text{(expand)}
\end{aligned}
$$

We define $r(t) = \frac{3t^2-4t}{(3t-2)^2}$ and set about constructing a sign diagram for r. Solving $(3t-2)^2 = 0$ gives $t = \frac{2}{3}$, our sole excluded value. To find the zeros of r, we set $r(t) = \frac{3t^2-4t}{(3t-2)^2} = 0$ and solve $3t^2 - 4t = 0$. Factoring gives

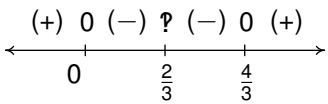


Figure 3.3.3

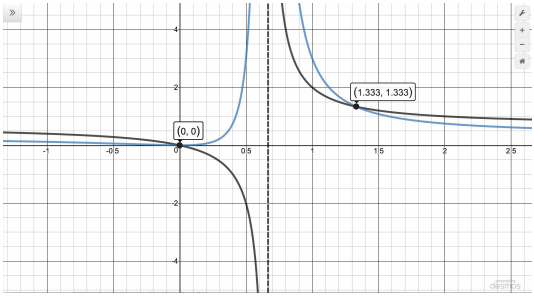


Figure 3.3.4

$t(3t-4) = 0$ so our solutions are $t = 0$ and $t = \frac{4}{3}$. After choosing test values, we get the sign diagram in Figure 3.3.3. Since we are looking for where $r(t) \leq 0$, we are looking for where $r(t)$ is $(-)$ or $r(t) = 0$. Hence, our final answer is $\left[0, \frac{2}{3}\right) \cup \left(\frac{2}{3}, \frac{4}{3}\right]$. In Figure 3.3.4, we graph $f(t) = 2t(3t-1)^{-1}$ (the darker curve), $g(t) = 3t^2(3t-2)^{-2}$, and vertical asymptote $x = \frac{2}{3}$, the dashed line. Sure enough, the graph of f intersects the graph of g when $t = 0$ and $t = \frac{4}{3}$. Moreover, the graph of f is below the graph of g everywhere they are defined between these values, in accordance with our algebraic solution.

□

One thing to note about Example 3.3.2 is that the quantity $(3t-2)^2 \geq 0$ for all values of t. Hence, as long as we remember $t = \frac{2}{3}$ is excluded from consideration, we could actually multiply both sides of the inequality in Example 3.3.2 by $(3t-2)^2$ to obtain $2t(3t-2) \leq 3t^2$. We could then solve this (slightly easier) inequality using the methods of Section **??** as long as we remember to exclude $t = \frac{2}{3}$ from our solution. Once again, the more you *understand*, the less you have to *memorize*. If you know the

'*why*' behind an algorithm instead of just the '*how*,' you will know when you can short-cut it.

Our next example is an application of average cost. Recall from Definition **??** if $C(x)$ represents the cost to make x items then the average cost per item is given by $\overline{C}(x) = \frac{C(x)}{x}$, for $x > 0$.

Example 3.3.3. Recall from Example **??** that the cost, $C(x)$, in dollars, to produce x PortaBoy game systems for a local retailer is $C(x) = 80x + 150$, $x \geq 0$.

1. Find an expression for the average cost function, $\overline{C}(x)$.
2. Solve $\overline{C}(x) < 100$ and interpret.
3. Determine the behavior of $\overline{C}(x)$ as $x \to \infty$ and interpret.

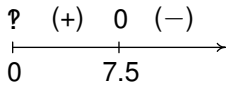
Solution.

1. From $\overline{C}(x) = \frac{C(x)}{x}$, we obtain $\overline{C}(x) = \frac{80x+150}{x}$. The domain of C is $x \geq 0$, but since $x = 0$ causes problems for $\overline{C}(x)$, we get our domain to be $x > 0$, or $(0, \infty)$.
2. Solving $\overline{C}(x) < 100$ means we solve $\frac{80x+150}{x} < 100$. We proceed as in the previous example.

$$\begin{aligned} \frac{80x + 150}{x} &< 100 \\ \frac{80x + 150}{x} - 100 &< 0 \\ \frac{80x + 150 - 100x}{x} &< 0 && \text{(common denominator)} \\ \frac{150 - 20x}{x} &< 0 \end{aligned}$$

If we take the left hand side to be a rational function $r(x)$, we need to keep in mind that the applied domain of the problem is $x > 0$.

This means we consider only the positive half of the number line for our sign diagram. On $(0, \infty)$, r is defined everywhere so we need only look for zeros of r. Setting $r(x) = 0$ gives $150 - 20x = 0$, so that $x = \frac{15}{2} = 7.5$. The test intervals on our domain are $(0, 7.5)$ and $(7.5, \infty)$. We find $r(x) < 0$ on $(7.5, \infty)$.



In the context of the problem, x represents the number of PortaBoy games systems produced and $\overline{C}(x)$ is the average cost to produce each system. Solving $\overline{C}(x) < 100$ means we are trying to find how many systems we need to produce so that the average cost is less than \$100 per system. Our solution, $(7.5, \infty)$ tells us that we need to produce more than 7.5 systems to achieve this. Since it doesn't make sense to produce half a system, our final answer is $[8, \infty)$.

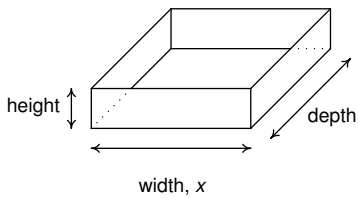
3. When we apply Theorem **??** to $\overline{C}(x)$ we find that $y = 80$ is a horizontal asymptote to the graph of $y = \overline{C}(x)$. To more precisely determine the behavior of $\overline{C}(x)$ as $x \to \infty$, we first use long division[2] and rewrite $\overline{C}(x) = 80 + \frac{150}{x}$. As $x \to \infty$, $\frac{150}{x} \to 0^+$, which means $\overline{C}(x) \approx 80 +$very small $(+)$. Thus the average cost per system is getting closer to \$80 per system. If we set $\overline{C}(x) = 80$, we get $\frac{150}{x} = 0$, which is impossible, so we conclude that $\overline{C}(x) > 80$ for all $x > 0$. This means that the average cost per system is always greater than \$80 per system, but the average cost is approaching this amount as more and more systems are produced. Looking back at Example **??**, we realize \$80 is the variable cost per system $-$ the cost per system above and beyond the fixed initial cost of \$150. Another way to interpret our answer is that 'infinitely' many systems would need to be produced to effectively 'zero out' the fixed cost. □

Note that number 2 in Example 3.3.3 is another opportunity to short-cut the standard algorithm and obtain the solution more quickly if we take stock of the situation. Since the applied domain is $x > 0$, we can multiply

[2] In this case, long division amounts to term-by-term division.

through the inequality $\frac{80x+150}{x} < 100$ by x without worrying about changing the sense of the inequality. This reduces the problem to $80x+150 < 100x$, a basic linear inequality whose solution is readily seen to be $x > 7.5$. It is absolutely critical here that $x > 0$. Indeed, any time you decide to multiply an inequality by a variable expression, it is necessary to justify why the inequality is preserved. Our next example is another classic 'box with no top' problem. The reader is encouraged to compare and contrast this problem with Example **??** in Section **??**.

Example 3.3.4. A box with a square base and no top is to be constructed so that it has a volume of 1000 cubic centimeters. Let x denote the width of the box, in centimeters as seen below.



1. Explain why the height of the box (in centimeters) is a function of the width x. Call this function h and find an expression for $h(x)$, complete with an appropriate applied domain.
2. Solve $h(x) \geq x$ and interpret.
3. Find and interpret the behavior of $h(x)$ as $x \to 0^+$ and as $x \to \infty$.
4. Express the surface area of the box as a function of x, $S(x)$ and state the applied domain.
5. Use a graphing utility to approximate (to two decimal places) the dimensions of the box which minimize the surface area.

Solution.

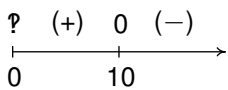
1. We are told that the volume of the box is 1000 cubic centimeters and that x represents the width, in centimeters. Since x represents a physical dimension of a box, we have that $x > 0$. From geometry, we know volume = width $\times$ height $\times$ depth. Since the base of the box

is a square, the width and the depth are both x centimeters. Hence, $1000 = x^2(\text{height})$. Solving for the height, we get height $= \frac{1000}{x^2}$. In other words, for each width $x > 0$, we are able to compute *the*[3] corresponding height using the formula $\frac{1000}{x^2}$. Hence, the height is a function of x. Using function notation, we write $h(x) = \frac{1000}{x^2}$. As mentioned before, our only restriction is $x > 0$ so the domain of h is $(0, \infty)$.

2. To solve $h(x) \geq x$, we proceed as before and collect all nonzero terms on one side of the inequality in order to use a sign diagram.

$$\begin{aligned} h(x) &\geq x \\ \frac{1000}{x^2} &\geq x \\ \frac{1000}{x^2} - x &\geq 0 \\ \frac{1000 - x^3}{x^2} &\geq 0 && \text{(common denominator)} \end{aligned}$$

We consider the left hand side of the inequality as our rational function $r(x)$. We see immediately the only value excluded from the domain of r is 0, but since our applied domain is $x > 0$, we restrict our attention to the interval $(0, \infty)$. The sole zero of r comes when $1000 - x^3 = 0$, or when $x = 10$. Choosing test values in the intervals $(0, 10)$ and $(10, \infty)$ gives the following:



We see $r(x) > 0$ on $(0, 10)$, and since $r(x) = 0$ at $x = 10$, our solution is $(0, 10]$. In the context of the problem, $h(x)$ represents the height of the box while x represents the width (and depth) of the

[3] that is, the one and only one

box. Solving $h(x) \geq x$ is tantamount to finding the values of x which result in a box where the height is at least as big as the width (and, in this case, depth.) Our answer tells us the width of the box can be at most 10 centimeters for this to happen.[4]

3. As $x \to 0^+$, $h(x) = \frac{1000}{x^2} \to \infty$. This means that the smaller the width x (and, in this case, depth), the larger the height h has to be in order to maintain a volume of 1000 cubic centimeters. As $x \to \infty$, we find $h(x) \to 0^+$, which means that in order to maintain a volume of 1000 cubic centimeters, the width and depth must get bigger as the height becomes smaller.

4. Since the box has no top, the surface area can be found by adding the area of each of the sides to the area of the base. The base is a square of dimensions x by x, and each side has dimensions x by $h(x)$. We get the surface area, $S(x) = x^2 + 4xh(x)$. Since $h(x) = \frac{1000}{x^2}$, we have $S(x) = x^2 + 4x\left(\frac{1000}{x^2}\right) = x^2 + \frac{4000}{x}$. The domain of S is the same as h, namely $(0, \infty)$, for the same reasons as above.

5. To graph $y = S(x)$, we create a table of values to help us define a good viewing window. Doing so, we find a local minim when $x \approx 12.60$. As far as we can tell,[5] this is the only local extremum, so it is the (absolute) minimum as well. This means that the width and depth of the box should each measure approximately 12.60 centimeters. To determine the height, we find $h(12.60) \approx 6.30$, so the height of the box should be approximately 6.30 centimeters.[6] See Figure 3.3.5

□

Our last example uses regression to verify a very famous scientific law.

[4] As with the previous example, knowing $x > 0$ means $x^2 > 0$ so we can clear denominators right away and solve $x^3 \leq 1000$, or $x \leq 10$. Coupled with our applied domain, $x > 0$, we would arrive at the same solution, $(0, 10]$.

[5] without Calculus, that is...

[6] The y-coordinate here, 476.22 means the minimum surface area possible is 476.22 square centimeters. Minimizing the surface area minimizes the material required to make the box, therein helping to reduce the cost of the box.

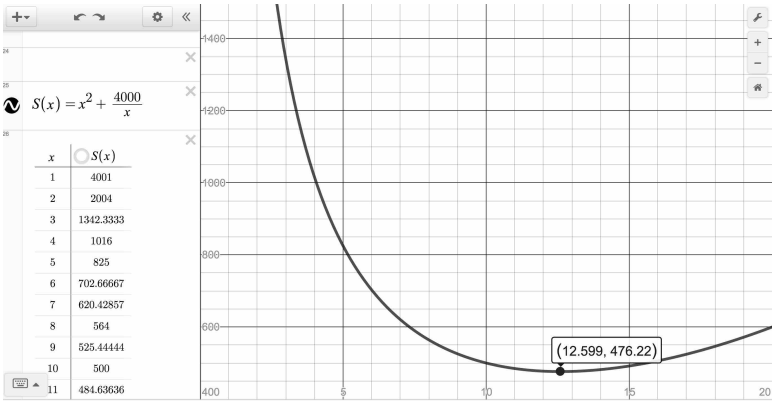


Figure 3.3.5

Example 3.3.5. Boyle's Law states that when temperature is held constant, the pressure of a gas is inversely proportional to the volume of the gas.[7] According to this website[8] the actual data relating the volume V of a gas and its pressure P used by Boyle and his assistant in 1662 to formulate this law is given in Table 3.3.1. (NOTE: both pressure and volume here are given in 'arbitrary units.')

1. Assuming P and V are inversely proportional, estimate the constant of proportionality, k.

2. Use a graphing utility to fit a curve of the form $P = \dfrac{k}{V}$ to these data.

Solution.

1. Recall if P and V are inversely proportional, there is a real number k so $PV = k$ for all values of P and V. Multiplying the corresponding P and V values from the data together result in numbers which are consistently approximately 1400. This gives us confidence in the claim P and V are inversely proportional and suggests $k \approx 1400$.

[7] For a review of what this means, see Section **??**.

[8] http://web.lemoyne.edu/~giunta/classicalcs/boyleverify.html

V	48	46	44	42	40	38	36	34	32
P	29.13	30.56	31.94	33.5	35.31	37	39.31	41.63	44.19

V	30	28	26	24	23	22	21	20
P	47.06	50.31	54.31	58.81	61.31	64.06	67.06	70.69

V	19	18	17	16	15	14	13	12
P	74.13	77.88	82.75	87.88	93.06	100.44	107.81	117.56

Table 3.3.1

2. We plot the pairs (V, P) and run a regression, the results of which are in Figure 3.3.6. To our amazement, the graphing utility reports $k \approx 1406.9$ with $R^2 \approx 1$. This means the data are a very good fit to the model $P = \frac{k}{V}$, or $PV = k$, hence verifying Boyle's Law for this set of data.

$\square$

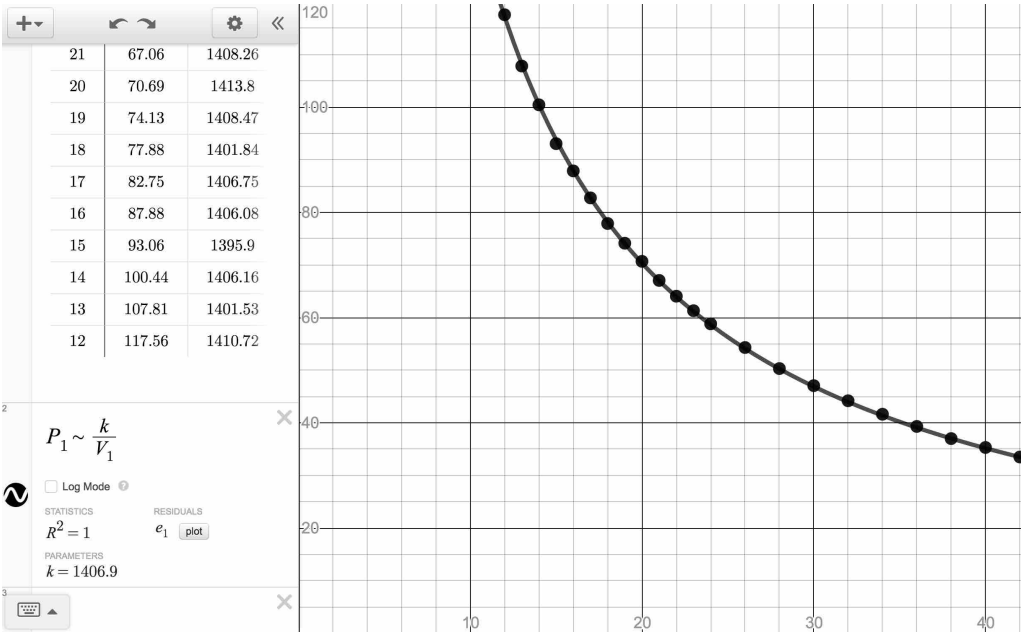

21 67.06 1408.26
20 70.69 1413.8
19 74.13 1408.47
18 77.88 1401.84
17 82.75 1406.75
16 87.88 1406.08
15 93.06 1395.9
14 100.44 1406.16
13 107.81 1401.53
12 117.56 1410.72
$P_1 \sim \frac{k}{V_1}$
Log Mode
STATISTICS
$R^2 = 1$
RESIDUALS
e_1
plot
PARAMETERS
$k = 1406.9$
120
100
80
60
40
20
10
20
30
40

Figure 3.3.6

3.3.1 Exercises

(Review of Solving Equations):[9] In Exercises 1 - 6, solve the rational equation. Be sure to check for extraneous solutions.

1. $\dfrac{x}{5x+4} = 3$
2. $\dfrac{3x-1}{x^2+1} = 1$
3. $\dfrac{1}{t+3} + \dfrac{1}{t-3} = \dfrac{t^2-3}{t^2-9}$
4. $\dfrac{2t+17}{t+1} = t+5$
5. $\dfrac{z^2-2z+1}{z^3+z^2-2z} = 1$
6. $\dfrac{4z-z^3}{z^2-9} = 4z$

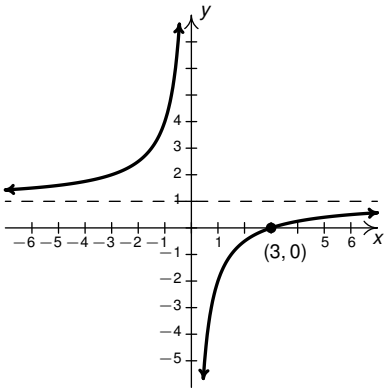
In Exercises 7 - 22, solve the rational inequality. Express your answer using interval notation.

7. $\dfrac{1}{x+2} \geq 0$
8. $\dfrac{5}{x+2} \geq 1$
9. $\dfrac{x}{x^2-1} < 0$
10. $\dfrac{4t}{t^2+4} \geq 0$
11. $\dfrac{2t+6}{t^2+t-6} < 1$
12. $\dfrac{5}{t-3} + 9 < \dfrac{20}{t+3}$
13. $\dfrac{6z+6}{2+z-z^2} \leq z+3$
14. $\dfrac{6}{z-1} + 1 > \dfrac{1}{z+1}$
15. $\dfrac{3z-1}{z^2+1} \leq 1$
16. $(2x+17)(x+1)^{-1} > x+5$
17. $(4x-x^3)(x^2-9)^{-1} \geq 4x$
18. $(x^2+1)^{-1} < 0$
19. $(2t-8)(t+1)^{-1} \leq (t^2-8t)(t+1)^{-2}$
20. $(t-3)(2t+7)(t^2+7t+6)^{-2} \geq (t^2+7t+6)^{-1}$
21. $60z^{-2} + 23z^{-1} \geq 7(z-4)^{-1}$
22. $2z + 6(z-1)^{-1} \geq 11 - 8(z+1)^{-1}$

In Exercises 23. - 28., use the the graph of the given rational function to solve the stated inequality.

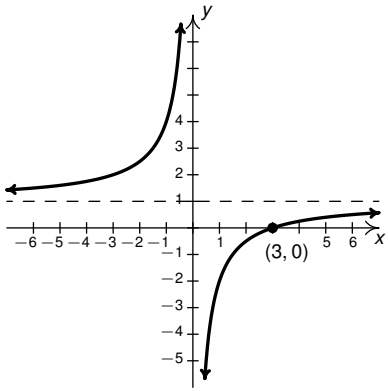
[9] For more review, see Section **??**.

23. Solve $f(x) \geq 0$.



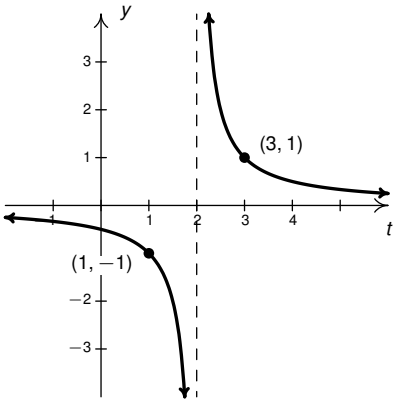
$y = f(x)$, asymptotes: $x = 0$, $y = 1$.

24. Solve $f(x) < 1$.



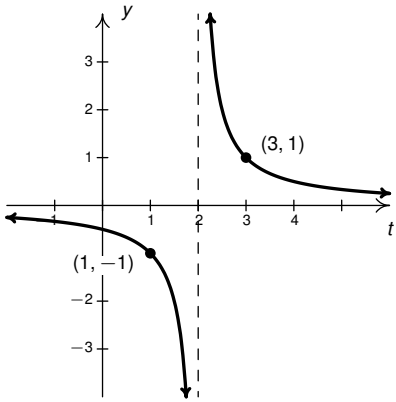
$y = f(x)$, asymptotes: $x = 0$, $y = 1$.

25. Solve $g(t) \geq -1$.



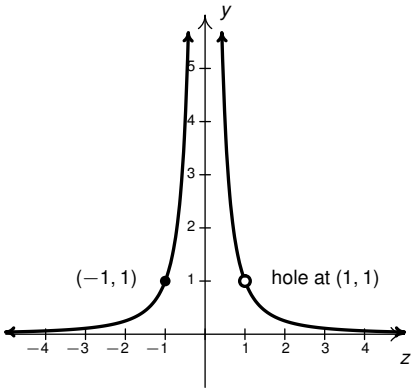
$y = g(t)$, asymptotes: $t = 2$, $y = 0$.

26. Solve $-1 \leq g(t) < 1$.



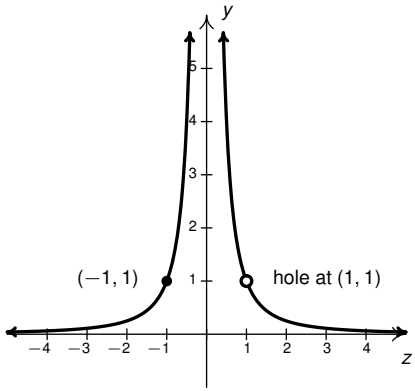
$y = g(t)$, asymptotes: $t = 2$, $y = 0$.

27. Solve $r(z) \leq 1$



$y = r(z)$, asymptotes: $z = 0$, $y = 0$.

28. Solve $r(z) > 0$.



$y = r(z)$, asymptotes: $z = 0$, $y = 0$.

29. In Exercise **??** in Section **??**, the function $C(x) = .03x^3 - 4.5x^2 + 225x + 250$, for $x \geq 0$ was used to model the cost (in dollars) to produce x PortaBoy game systems. Using this cost function, find the number of PortaBoys which should be produced to minimize the average cost $\overline{C}$. Round your answer to the nearest number of systems.

30. Suppose we are in the same situation as Example 3.3.4. If the volume of the box is to be 500 cubic centimeters, use a graphing utility to find the dimensions of the box which minimize the surface area. What is the minimum surface area? Round your answers to two decimal places.

31. The box for the new Sasquatch-themed cereal, 'Crypt-Os', is to have a volume of 140 cubic inches. For aesthetic reasons, the height of the box needs to be 1.62 times the width of the base of the box.[10] Find the dimensions of the box which will minimize the surface area of the box. What is the minimum surface area? Round your answers to two decimal places.

32. Sally is Skippy's neighbor from Exercise **??** in Section **??**. Sally also wants to plant a vegetable garden along the side of her home. She

[10] 1.62 is a crude approximation of the so-called 'Golden Ratio' $\phi = \frac{1+\sqrt{5}}{2}$.

doesn't have any fencing, but wants to keep the size of the garden to 100 square feet. What are the dimensions of the garden which will minimize the amount of fencing she needs to buy? What is the minimum amount of fencing she needs to buy? Round your answers to the nearest foot. (Note: Since one side of the garden will border the house, Sally doesn't need fencing along that side.)

33. Another Classic Problem: A can is made in the shape of a right circular cylinder and is to hold one pint. (For dry goods, one pint is equal to 33.6 cubic inches.)[11]

    (a) Find an expression for the volume V of the can in terms of the height h and the base radius r.

    (b) Find an expression for the surface area S of the can in terms of the height h and the base radius r. (Hint: The top and bottom of the can are circles of radius r and the side of the can is really just a rectangle that has been bent into a cylinder.)

    (c) Using the fact that $V = 33.6$, write S as a function of r and state its applied domain.

    (d) Use your graphing calculator to find the dimensions of the can which has minimal surface area.

34. A right cylindrical drum is to hold 7.35 cubic feet of liquid. Find the dimensions (radius of the base and height) of the drum which would minimize the surface area. What is the minimum surface area? Round your answers to two decimal places.

35. In Exercise **??** in Section **??**, the population of Sasquatch in Portage County is modeled by

$$P(t) = \frac{150t}{t+15}, \quad t \geq 0,$$

where $t = 0$ corresponds to the year 1803. According to this model, when were there fewer than 100 Sasquatch in Portage County?

[11] According to www.dictionary.com[12], there are different values given for this conversion. We use 33.6in^3 for this problem.

3.3.2 Answers

1. $x = -\frac{6}{7}$
2. $x = 1,\ x = 2$
3. $t = -1$
4. $t = -6,\ x = 2$
5. No solution
6. $z = 0,\ z = \pm 2\sqrt{2}$
7. $(-2, \infty)$
8. $(-2, 3]$
9. $(-\infty, -1) \cup (0, 1)$
10. $[0, \infty)$
11. $(-\infty, -3) \cup (-3, 2) \cup (4, \infty)$
12. $\left(-3, -\frac{1}{3}\right) \cup (2, 3)$
13. $(-1, 0] \cup (2, \infty)$
14. $(-\infty, -3) \cup (-2, -1) \cup (1, \infty)$
15. $(-\infty, 1] \cup [2, \infty)$
16. $(-\infty, -6) \cup (-1, 2)$
17. $(-\infty, -3) \cup \left[-2\sqrt{2}, 0\right] \cup \left[2\sqrt{2}, 3\right)$
18. No solution
19. $[-4, -1) \cup (-1, 2]$
20. $(-\infty, -6) \cup (-6, -3] \cup [9, \infty)$
21. $[-3, 0) \cup (0, 4) \cup [5, \infty)$
22. $\left(-1, -\frac{1}{2}\right] \cup (1, \infty)$
23. $f(x) \geq 0$ on $(-\infty, 0) \cup [3, \infty)$.
24. $f(x) < 1$ on $(0, \infty)$.
25. $g(t) \geq -1$ on $(-\infty, 1] \cup (2, \infty)$.
26. $-1 \leq g(t) < 1$ on $(-\infty, 1] \cup (3, \infty)$.
27. $r(z) \leq 1$ on $(-\infty, -1] \cup (1, \infty)$.
28. $r(z) > 0$ on $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$.
29. The absolute minimum of $y = \overline{C}(x)$ occurs at $\approx (75.73, 59.57)$. Since x represents the number of game systems, we check $\overline{C}(75) \approx 59.58$ and $\overline{C}(76) \approx 59.57$. Hence, to minimize the average cost, 76 systems should be produced at an average cost of \$59.57 per system.
30. The width (and depth) should be 10.00 centimeters, the height should be 5.00 centimeters. The minimum surface area is 300.00 square centimeters.
31. The width of the base of the box should be approximately 4.12 inches, the height of the box should be approximately 6.67 inches, and the depth of the base of the box should be approximately 5.09 inches. The minimum surface area is approximately 164.91 square inches.
32. The dimensions are approximately 7 feet by 14 feet. Hence, the

minimum amount of fencing required is approximately 28 feet.

33. (a) $V = \pi r^2 h$ (b) $S = 2\pi r^2 + 2\pi rh$
    (c) $S(r) = 2\pi r^2 + \frac{67.2}{r}$, Domain $r > 0$
    (d) $r \approx 1.749$ in. and $h \approx 3.498$ in.

34. The radius of the drum should be approximately 1.05 feet and the height of the drum should be approximately 2.12 feet. The minimum surface area of the drum is approximately 20.93 cubic feet.

35. $P(t) < 100$ on $(-15, 30)$, and the portion of this which lies in the applied domain is $[0, 30)$. Since $t = 0$ corresponds to the year 1803, from 1803 through the end of 1832, there were fewer than 100 Sasquatch in Portage County.